

地质学专业（基地班）培养方案

专业名称与代码：070901

专业培养目标：本专业旨在培养面向未来国家发展重大需求，具有优秀的政治素质、道德品质和科学文化素养，适应全球化和信息化时代发展的地球科学领域高素质人才。毕业生需具备深厚的人文素养、坚实的数理化学基础知识和数据处理技能，掌握较全面的地球系统科学理论和技术方法，具有扎实的实践能力和较强的科研创新能力，可满足 21 世纪国家在资源、环境、灾害、国土规划与国民经济其它相关领域的高端人才需求。

专业毕业要求：

1. 思想道德与人文素养：具有坚定的政治立场、良好的道德修养和正确的人生观与价值观；具有家国情怀和社会责任感，具备艰苦朴素、求真务实的地质专业精神；具有一定的人文和艺术修养。

2. 理科基础与思维能力：具有扎实的数学、物理、化学、生物学和信息科学基础，具有较强的逻辑思维能力和分析解决问题的能力。

3. 专业知识与技能：掌握地球物质、地球化学、地球物理学、地球观测、地球表层系统、地球环境与生命演化、行星科学等领域基础专业知识体系，具备地球系统科学思维；具有在地球科学各领域从事相关工作的基本知识和技能。

4. 科研与实践能力：掌握地球科学研究的基本方法和常用技术手段，具有较强的野外地质工作技能；掌握至少一门计算机语言的编程技术，能熟练地进行信息获取、处理和表达，具有大数据获取及分析的基本技能，具备从事科学研究的能力和开展科研活动的经历。

5. 国际视野与交流能力：了解地质学科国际前沿发展趋势，具有较流利的外语听、说、读、写能力，具备一定的国际视野和国际交流能力；具有较好的组织管理能力和团队合作能力。

6. 自主学习与创新能力：具有较强的自主学习能力，能够根据自身发展和社会需求积极开展自我完善和自主学习；在了解学科前沿、国家目标及国际形势的基础上，积极开展创新性实践与探索。

毕业要求及实现途径：

序号	毕业要求	实现途径（教学过程）
1	思想道德与人文素养： 具有坚定的政治立场、良好的道德修养和正确的人生观与价值观；具有家国情怀和社会责任感，具备艰苦朴素、求真务实的地质专业精神；具有一定的人文和艺术修养。	①课堂教学： 思想道德修养与法律基础、马克思主义基本原理、毛泽东思想和中国特色社会主义理论体系、习近平新时代中国特色社会主义思想概论、体育、军事理论、大学语文、传统文化及中西方文化选修课、哲学基础、人文艺术类选修课 ②课外学习： 经典阅读与分享、人文讲座、网上主题团课、社会调查等
2	理科基础与思维能力： 具有较好的数学、物理、化学、生物学和信息科学基础，具有较强的逻辑思维能力和分析解决问题的能力。	①课堂教学： 高等数学及地学应用、线性代数及地学应用、概率论与数理统计及地学应用、大学物理及地学应用、物理实验、普通生物学、大学化学及地学应用、化学实验、逻辑学等 ②课外学习： 大学生数学建模、物理实验创新设计、化学实验创新设计等竞赛活动及相关的系列科普活动
3	专业知识与技能： 掌握地球物质、地球化学、地球物理学、地球观测、地球表层系统、地球环境与生命演化、行星科学等领域基础专业知识体系，具备地球系统科学思维；具有在地球科学各领域从事相关工作的基本知识和技能。	①课堂教学： 地球系统科学导论、地球化学、地球物理学概论等学科基础课；地球物质、地球表层系统科学、深时地球环境与生命过程等专业核心课；大气科学概论、海洋学基础、环境科学导论、经济地理学、地球物理与空间信息导论、环境规划与管理等跨学科/专业选修课；北戴河和周口店野外地质实习、海内外经典地质路线实习 ②课外学习： “地球系统科学论坛”、“名家论坛”等系列学术报告与交流
4	科研与实践能力： 掌握地球科学研究的基本方法和常用技术手段，具有较强的野外地质工作技能；掌握至少一门计算机语言的编程技术，能熟练地进行信息获取、处理和表达，具有大数据获取及分析的基本技能，具备从事科学研究的能力和开展科研活动的经历。	①课堂教学： 北戴河、周口店和秭归三大野外地质实习；海内外经典地质路线实习；地球物质、地球生物学、构造地质学等专业课实习；科技论文写作、Python 语言程序课程设计、人工智能等相关课程 ②课外学习： 参观国家重点实验室；二年级进入导师科研团队，参与科研项目工作，三年级开始自由组队开始创新课题研究；参加大学生科技论文报告会、大学生创新创业训练计划项目
5	国际视野与交流能力： 了解地质学科国际前沿发展趋势，具有较流利的外语听、说、读、写能力，具备一定的国际视野和国际交流能力；具有较好的	①课堂教学： 大学英语、部分学科基础课和专业核心课全英文教学、海外经典地质路线实习等 ②课外学习： 外文文献阅读、大学生英语竞赛、国外名家学术讲

序号	毕 业 要 求	实现途径（教学过程）
	组织管理能力和团队合作能力。	座、与本校留学生交流活动、世界名校访学交流等
6	自主学习与创新意识： 具有较强的自主学习能力，能够根据自身发展和社会需求积极开展自我完善和自主学习；在了解学科前沿、国家目标及国际形势的基础上，积极开展创新性实践与探索。	①课堂教学： 以问题为导向的专业课程教学、以学生为主体的研讨式课堂教学 ②课外学习： 团队化科研训练、实践能力训练、社会调查、“赛恩斯”科技领航系列活动、“互联网+”、“挑战杯”等重大科创赛事

主干学科：地质学

专业核心课程：地球系统科学导论、地球物质科学、地球生物学、沉积地球系统科学、构造地质学、地貌学与第四纪地质学、地球化学、地球物理学概论、矿床学

主要专业实验：晶体模型定向与分析、矿物标本与镜下鉴定、岩石手标本与薄片鉴定、古生物鉴定、地质图读图与构造分析、地层与地史学实习

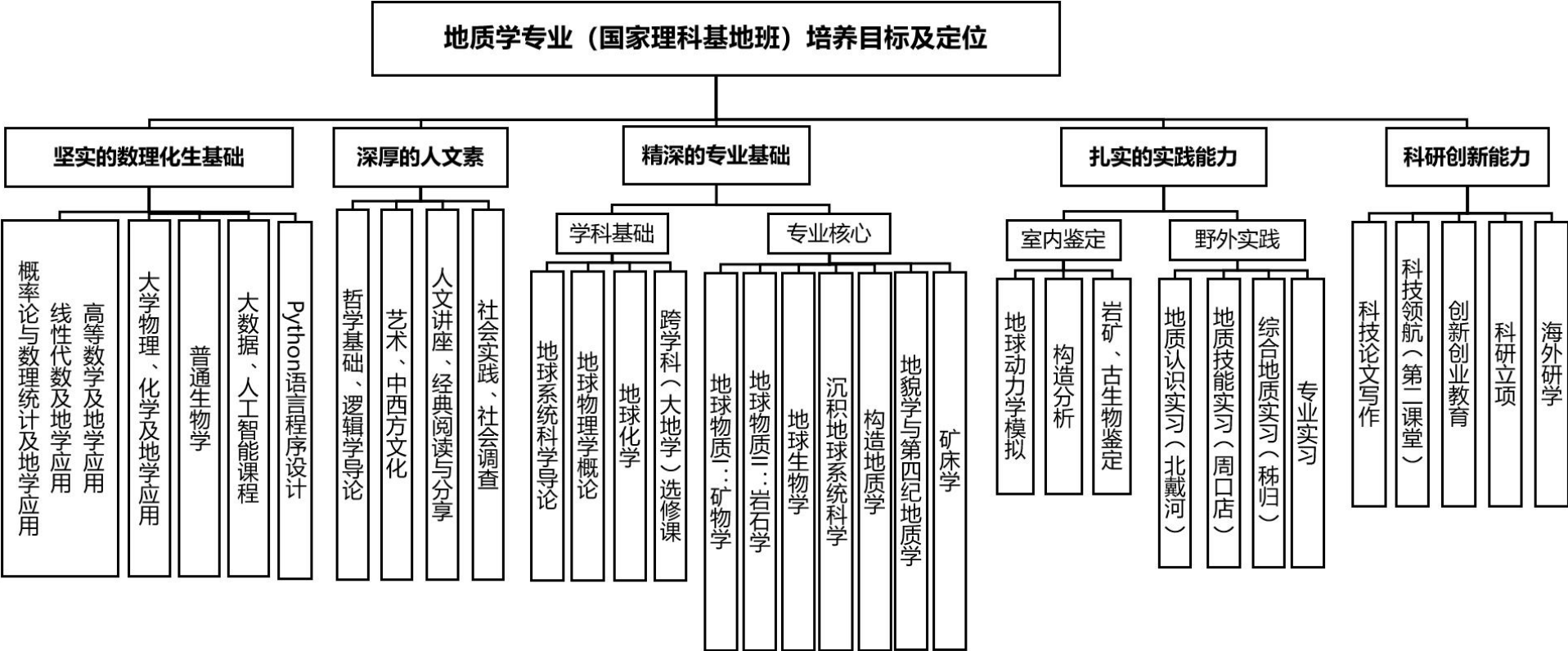
主要实践性教学环节：地质认识实习（北戴河）、地质技能实习（周口店）、综合地质实习（秭归）、毕业生产实习、毕业论文

毕业学分要求：161.5 学分

学制与学位：四年，理学学士

本专业学生可以辅修的其他专业：地理科学、大气科学、海洋科学、地质工程、环境工程

相近专业：地球化学、地球物理学



Program for Geology Major

(National Training Base for Geology)

Specialty and Code: Geology 070901

Education Objective: This major aims to cultivate high-quality talents in the field of geoscience with strong political stance, excellent moral quality and scientific and cultural literacy to meet the future needs of national development and adapt to the development of globalization and future technologies. Graduates should have profound humanistic quality, solid basic knowledge of mathematics, physics and chemistry, data processing skills, comprehensive theory and technical methods of earth system science, solid practical ability and strong scientific research and innovation ability. Over 70% of the graduates will continue to study for a master's degree or a master's and a doctoral degree. Most of the graduates can become senior talents engaged in research or management in geoscience and related fields in scientific research institutions and universities, which can meet the needs of high-level talents in resources, environment, geohazards, land planning and other related fields of national economy in the 21st century.

Graduation Requirements:

1. Moral & humanistic quality: Have a firm political stance and a strong sense of social responsibility; have a correct outlook on life and values; possess a professional spirit of hard work, simplicity, and pragmatism.

2. Scientific literacy and critical thinking: Have a solid foundation in mathematics, physics, chemistry, biology and information science and the ability to analyze and solve problems, with strong logical thinking ability.

3. Professional foundation: Establish a basic knowledge system of Earth materials, geochemistry, geophysics, Earth observation, Earth surface system, Earth environment and life evolution, and have the vision of Earth system science; have the basic knowledge and skills to engage in related work in various fields of geosciences.

4. Research and practical ability: Master the basic methods and common technical means for solving scientific problems in the field of Earth sciences; have strong skills in field geological work; master the programming techniques of at least one computer language, be able to skillfully acquire, process and express information; have the basic skills of acquiring and analyzing big data; have the ability to engage in scientific research and the experience of

carrying out scientific research activities.

5. International Vision and Intercultural Communication: Understand the development trend of geosciences; have fluent listening, speaking, reading and writing skills in a foreign language; have an international vision and intercultural communication ability; have a certain degree of organization and management ability and team cooperation ability.

6. Independent Learning and Creativity: Be able to actively engage in independent study in accordance with their own development and the needs of society; to actively engage in innovative practice and exploration on the basis of an understanding of the frontiers of geosciences, national goals and the international situation.

Graduation Requirements and Pathways:

No	Graduation Requirements	Pathways (Teaching Process)
1	Moral & humanistic quality: Have a firm political stance and a strong sense of social responsibility; have a correct outlook on life and values; possess a professional spirit of hard work, simplicity, and pragmatism.	①Classroom Teaching: Ideological and Moral Cultivation and legal Foundation, Situation and Policy, Introduction to Mao Zedong Thought and the Theoretical System of Socialism with Chinese Characteristics, Labor Practice, College Chinese, Traditional Culture and Chinese and Western Culture Elective Courses, Philosophy Foundation, Aesthetics and Art Elective Courses ②Extracurricular Learning: Classic reading and sharing, humanities lectures, social surveys, etc.
2	Scientific literacy and critical thinking: Have a solid foundation in mathematics, physics, chemistry, biology and information science and the ability to analyze and solve problems, with strong logical thinking ability.	①Classroom Teaching: Advanced Mathematics and Geoscience Application, Linear Algebra and Geoscience Application, Probability Theory and Mathematical Statistics and Geosciences Application, College Physics and Geosciences Application, Physics Experiment, General Biology, College Chemistry and Geoscience Application, Chemistry Experiment, Logic, etc. ②Extracurricular Learning: Competition activities such as Mathematical Modeling, Innovative Design of Physical Experiments, Innovative Design of Chemical Experiments, and a series of popular science activities
3	Professional foundation: Establish a basic knowledge system of Earth materials, geochemistry, geophysics, Earth observation, Earth surface system, Earth environment and life evolution, and have the vision of Earth system science; have the basic knowledge and skills to engage in related work in various fields of	①Classroom Teaching: Introduction to Earth System Science (basic course); major core courses such as Earth Surface System Science, Deep Earth Environment and Life Process, and Deep Earth Science; major elective courses such as Introduction to Atmospheric Science, Fundamentals of Oceanography, Introduction to Environmental Science, Economic Geography,

No	Graduation Requirements	Pathways (Teaching Process)
	geosciences.	Introduction to Geophysics and Spatial Information, and Environmental Planning and Management; Field geological practice in Beidaihe and Zhoukoudian, and classic geological route practice at home and abroad ②Extracurricular Learning: Series of academic reports and exchanges such as "Earth System Science Forum" and "Famous Experts Forum"
4	Research and practical ability: Master the basic methods and common technical means for solving scientific problems in the field of Earth sciences; have strong skills in field geological work; master the programming techniques of at least one computer language, be able to skillfully acquire, process and express information; have the basic skills of acquiring and analyzing big data; have the ability to engage in scientific research and the experience of carrying out scientific research activities.	①Classroom Teaching: Beidaihe and Zhoukoudian Field Geological Practice, Classic Geological Route Practice at Home and Abroad, Scientific Paper Writing, Python Language Program Course Design, Artificial Intelligence and other related courses ②Extracurricular Learning: Scientific research projects: joining the tutor's scientific research team and participated in scientific research projects, organizing teams freely to start innovative research projects, and participate in 1-2 academic conferences; college students' scientific and technological thesis reports, and college students' innovation and entrepreneurship training programs.
5	International Vision and Intercultural Communication: Understand the development trend of geosciences; have fluent listening, speaking, reading and writing skills in a foreign language; have an international vision and intercultural communication ability; have a certain degree of organization and management ability and team cooperation ability.	①Classroom Teaching: College English, bilingual teaching of selected specialty courses ②Extracurricular Learning: Literature reading in English, seminars by English-speaking experts, interactions with international students, visits and exchanges at famous overseas universities
6	Independent Learning and Creativity: Be able to actively engage in independent study in accordance with their own development and the needs of society; to actively engage in innovative practice and exploration on the basis of an understanding of the frontiers of geosciences, national goals and the international situation.	①Classroom Teaching: Problem-oriented classroom teaching, student-centered seminar discussion ②Extracurricular Learning: Scientific research training, practical ability training, social investigation, discipline competition, academic report and discussion, etc.

Major Disciplines: Geology

Main Courses: Introduction to Earth System Science, Earth Materials (I, II), Geobiology, Sedimentary Earth System Science, Structural Geology, Geomorphology and Quaternary Geology, Geochemistry, Geophysics, Mineral Deposits.

Lab Experiments: Identification of Crystal Models and Mineral, Rock Identification, Fossil Identification, Historical Geology, Tectonic Analysis.

Practical Work: Primary Field Training (Beidaihe), Geological Mapping Training (Zhoukoudian in Beijing), Geological Field Training (Zigui), Professional Fieldwork Training, Practice for Graduation, Thesis for Graduation.

Credits Required for Graduation: 161.5

Duration& Degree Granted: Four years, Bachelor of Science

Recommended Minors: Geography Science, Geological Engineering, Atmospheric Science, Marine Science, Environmental Engineering

Related Specialties: Geochemistry, Geophysics

地质学专业（基地班）课程教学计划表

Course Descriptions of Geology Major Program
(National Training Base for Geology)

课程类别 Category		课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits							
						课内学时		课外学时				一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
						讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp									
通识教育课 Liberal Education Courses	必修 Compulsory	12007800	马克思主义基本原理 Principles of Marxism	3	48	48						3							
		12008100	毛泽东思想和中国特色社会主义理论体系概论 Introduction to Mao Tse-tung Thought & the Theoretical System of Socialism with Chinese Characteristics	2	32	32									2				
		12008000	习近平新时代中国特色社会主义思想概论 Introduction to Xi Jinping Thought on Socialism with Chinese Characteristics for a New Era	3	48	48									3				
		1171800	中国近现代史纲要 The Essentials of Modern Chinese History	2	32	32							2						
		12007900	思想道德与法治 Ideological morality & rule of law	3	48	32	16						3						
		12005300	形势与政策 Situation & Policy	2	32	32						每学期平均分配							
		113076*0	体育 Physical Education	4	144	144						1	1	1	1				
		109234*0	大学英语 College English	9	144	144				48		3	3	3					
		14300300	军事理论 Military Theory	2	36	36						2							
		12008200	劳动教育 Labor Education	1	16	16						1							
		21624800	大学语文 B College Chinese B	2	32	32							2						
		20123000	逻辑学导论 Logic	2	32	32						2							

地质学专业（基地班）

课程类别 Category		课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits							
						课内学时		课外学时				一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
						讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp									
		2180 1000	哲学基础 Foundation of Philosophy	2	32	32						2							
	选修 Elective	总计 12 学分。包括经典阅读与分享、人文讲座必修课程（各 1 学分），美育、心理健康教育课程（各不少于 2 学分），跨学科选修课不低于 4 学分。		12	192	192						4					4	4	
			人文讲座（1-7 学期） Humanities Seminars	1					16									1	
			经典阅读与分享（1-7 学期） Literature Reading & Sharing	1					16									1	
		小计 Sum		49	868	852	16		32	48		21	8	4	6	0	4	5	1
理科基础 Fundamentals of Science		2194 7400	Python 语言程序设计 Python Language Programming	2.5	40	40		16						2.5					
		2122 09*0	高等数学及地学应用 Advanced Mathematics & Application in Geosciences	11.5	184	184							5	6.5					
		2122 1000	线性代数及地学应用 Linear Algebra & Application in Geosciences	3.5	56	56									3.5				
		2122 0600	概率论与数理统计及地学应用 Probability Theory, Mathematical Statistics & Application in Geosciences	3.5	56	56										3.5			
		2122 11*0	大学物理及地学应用 College Physics & Application in Geosciences	8	128	128								4	4				
		2121 69*1	物理实验 A Physical Experiments A	2	64	4	60							1	1				
		2032 69*1	大学化学 A College Chemistry A	5.5	88	88								2.5	3				
		2032 70*1	大学化学实验 A College Chemistry Experiment A	2	48		48							1	1				
		2012 4400	科技论文写作 Science & Technology Paper Writing	1	16	16											1		
			小计 Sum		39.5	680	572	108	16					5	15	11.5	3.5	4.5	0

课程类别 Category		课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits							
						课内学时		课外学时				一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
						讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp									
学科内课程	20123500	地球系统科学导论 Introduction to Earth System Science	3	48	32	16			16		3								
	20120200	地球化学 Geochemistry	3	48	48				16	地球物质，构造地质学，沉积地球系统科学				3					
	20628000	地球物理学概论 Introduction of Geophysics	3	48	44	4				高数，大学物理，线性代数后面					3				
	20406900	普通生物学 General Biology	2	32	32						2								
学科基础课 Disciplinary Fundamental Courses	在学术导师建议下选择不少于 6 个学分的跨学科选修课																		
	22303700	人工智能 Artificial Intelligence	2	32	28	4										2			
	21130402	测量学 B Surveying B	1.5	24	24		12				1.5								
	21148500	地图学 Cartography	2	32	32		8								2				
	20111601	自然地理学 A Physical Geography A	3	48	48									3					
	20117700	遥感地质学 Remote Sense Geology	3	48	32	16									3				
	20202800	海洋学基础 Basic Oceanography	3	48	48	0										3			
	20121600	大气科学概论 Introduction to Atmospheric Science	1.5	24	24	0											1.5		
	21100700	GIS 原理与应用 Principle & Application of GIS	2.5	40	24	16											2.5		
	20539002	工程地质学基础 B Engineering Geology B	2	32	32	0												2	
	20222303	石油及天然气地质学 C Petroleum Geology C	2.5	40	40		8									2.5			
	20230802	矿产勘查理论与方法 B Theories & Methods of Mineral Resource Exploration B	2.5	40	40		8										2.5		
	20438900	环境评价 Environmental Assessment C	2	48	32					16								2	
	21704500	环境法规 Environmental Law	1.5	24	24													1.5	

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课程类别 Category		课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits							
						课内学时		课外学时				一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
						讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp									
		2050 6000	地质灾害防治 Geological Hazards Control	2	32	32					工程地质基础						2		
		2193 4300	地学大数据分析与应用 Big Data Analysis & Application	2	32	32								2					
		2192 4300	云计算与大数据处理技术 Cloud Computing & Big Data Processing	2	32	32					计算机网络 B							2	
		2044 2100	大气数值模拟 Atmospheric Numerical Simulation	1	16	16					大气科学概论							2	
		2083 2900	资源与环境经济学 Resource and Environmental Economics	2	32	32											2		
		2054 9200	灾害地质学 Geohazard Science	2	32	32												2	
		2040 4400	环境规划与管理 Environment Planning and Management	2	32	32												2	
		2011 3200	区域分析与规划学 Regional Analysis and Planning	2	32	32												2	
		小计 Sum		17	272	260	12			16	16		5	0	2	3	3	2	2
专业核心课程 Specialty Core Courses		2012 3710	地球物质 I: 矿物学 Earth Materials I: Mineralogy	4	64	28	36	4			大学化学		4						
		2012 3720	地球物质 II: 岩石学 Earth Materials II: Petrology	4	64	30	34	4	4		地球物质 I			4					
		2012 3800	地球生物学 Geobiology	3	48	32	16		8	8	地球系统科学导论			3					
		2012 3900	沉积地球系统科学 Sedimentary Earth System Science	4	64	48	16		8	8	地球生物学、地球物质 I, II、构造地质学				4				
		2010 4001	构造地质学 A Structural Geology A	4	64	40	24				地球系统科学导论、地球物质 II				4				
		2012 4500	地貌学与第四纪地质学 Geomorphology & Quaternary Geology	2.5	40	30	10			8						2.5			

课程类别 Category	课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits							
					课内学时		课外学时				一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
					讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp									
	2022 2502	矿床学 B Mineral Deposits B	2	32	32		16			地球物质 II					2			
	小计 Sum		23.5	376	240	136	24	20	24			4	7	8	4.5			
专业选修课 Specialty Elective Courses	在学术导师建议下选择不少于 6 个学分的专业选修课.																	
	2011 6300	区域地质调查技术与方法 Methods & Techniques in Regional Geological Survey	2	32	16	16										2		
	2011 3800	全球变化 A Global Change A	3	48	40	8									3			
	2012 4200	行星地质学导论 Introduction to Planetary Geology	2	32	20	12							2					
	2012 4800	应用矿物学 Applied Mineralogy	3	48	32	16									3			
	2012 4600	岩石学前沿与实践 Frontier & Practice of Petrology	3	48	24	24	16							3				
	2012 4900	应用地球化学 Applied Geochemistry	2	32	32					地球化学						2		
	2011 8900	地震地质学 Earthquake Geology	3	48	48										3			
	2012 5000	地质学虚拟现实方法与应用 Virtual Reality Methods & Applications in Geology	2	32	16	16										2		
	2012 5300	行星天文学 Planetary Astronomy	2	32	32												2	
	2012 5100	遥感数字图像处理原理与应用 Principles & Applications of Remote Sensing Digital Image Processing	2	32												2		
	2012 4300	古气候模拟 Paleoclimate simulations	2	32										2				
	4012 4700	地球生物学前沿实验技术方法与 实践 Methods & Practices of Analytical Techniques in Geobiology	1	1 周		1 周									1			

地质学专业（基地班）

课程类别 Category	课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits							
					课内学时		课外学时				一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
					讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp									
	40125400	海内外经典地质路线实习 Classic Geological Route Investigation	2	2 周														2
	小计 Sum		6	96	96									2	2	2		
合计 Sub-total			135	2292	2020	272	40	68	88		31	27	24.5	20.5	14	8	9	1
实践环节 Practical Work	44300400	军事训练 Military Training	2	2 周							2							
	42008300	劳动实践 Labor practice	1	4 周												1		
	41947500	Python 语言课程设计 Course Design for Python Language	1.5	1.5 周									1.5					
	40115200	地质认识实习（北戴河） Primary Field Training in Beidaihe	2	2 周								2						
	40124000	地质技能实习（周口店） Geological Field training in Zhoukoudian	4	4 周										4				
	40125202	综合地质实习（秭归） Advanced Geological Field Training in Zigui	2	2 周												2		
	40118500	毕业生产实习 Practice for Graduation	4	4 周													4	
	40125500	毕业论文（设计） Thesis for Graduation	4	4 周														4
	小计 Sum		20.5	23.5							2	2	1.5	4	0	3	4	4
创新创业自主学习 Freedom study	ZZ35000S	社会调查 Social Investigation	2	2 周											2			
		科研立项 Scientific Research	2	4 周												2		
		其他(含创新创业课程、学科竞赛、发明创造、科研报告) Others (Start-up, Contest, Invention, Innovation & Research Presentation)	2	2 周														2
	小计 Sum		6	8 周							0	0	0	0	2	2	0	2

中国地质大学（武汉）本科培养方案

课程类别 Category	课程编号 Code	课程名称 Course Title	学分 Crs	课内总学时 Hrs	学时分类 Class Hours					先修课程 Prerequisite courses	学期学分分配 Semester Credits							
					课内学时		课外学时				一 1st	二 2nd	三 3rd	四 4th	五 5th	六 6th	七 7th	八 8th
					讲课 Lec.	课内实验 Lab	实验/科研实践 Lab/Res.	研讨 Dis	素质拓展 Exp									
总计 Total			161.5	2292+31.5周	2020	272	40	68	88		33	29	26	24.5	16	13	13	7

注：全英课程须在课程名称后打*标出，通识教育选修课学分未列入具体学期，学院须根据学校创新创业自主学习学分认定一览表制订实施细则。

地质学（基地班）专业课程分类统计

Course Classifications and Statistics of Geology Major Program (National Training Base for Geosciences)

课程类别 统计	通识教育课程 Liberal Education Courses		理科基础课 Platform Fundamental Cours	学科基础课 Disciplinary Fundamental Cours		专业主干(核心)课 Main Specialty Courses	专业选修课 Specialty Elective Courses	实践环节 Practical Work	创新创业自主学习 Freedom Study	学时总计 Total Hours	学分总计 Total Credits
	必修	选修	必修	必修	选修						
课内总学时/学分	676/37	192/12	680/39.5	176/11	96/6	376/23.5	96/6	23.5 周/20.5	8 周/6	2292 +31.5 周	161.5
课内总学时/学分	868/49		952/56.5								
学分所占比例	30.3%		35%			14.6%	3.7%	12.7%	3.7%		100%